Fast Solar Filament Segmentation with a 4.3 M-Parameter Student Network

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

Solar filaments are thin, low-contrast structures, and annotators rarely agree on where one ends. We submit a 4.28 M-parameter network for the Solar Filament Segmentation Challenge 2026. It reads a full 2048×2048 H-α image in one forward pass at 26.9 images per second on a single RTX 4070. That is 9.5 times faster than the 171 M four-model ensemble we built as an accuracy reference, at 40 times fewer parameters. On the 107 validation files the submitted student reaches a panoptic quality (PQ) of 0.4591, against 0.4773 for the ensemble. Both score 0.38 on the public leaderboard, which is reported in steps of 0.01. Most of the gain we found came from post-processing rather than from architecture. Separating the contour threshold from the rule that accepts an instance added 0.0158 PQ in nested cross-validation and moved our earlier submission from 0.35 to 0.37 with no retraining. Eight-view test-time augmentation added the final step to 0.38.

CCS Concepts

• **Computing methodologies** → **Image segmentation**; *Supervised learning*.

Keywords

solar filaments, instance segmentation, panoptic quality, knowledge distillation, inference speed

1 Introduction

Filaments are long, dark threads in the solar chromosphere. Seen in H-α they cross the disk as thin curves, often with barbs, and two experts asked to outline the same filament usually draw different shapes. The Solar Filament Segmentation Challenge 2026 [2] asks for instance masks on 707 full-disk GONG images from the MAGFiLO dataset [1]. Each image carries one to three expert annotations, and the score is panoptic quality computed against every annotator, with a match at IoU> 0.5.

Our entry is a small network. It takes one full-disk image, produces a probability map, and turns that map into instances with a fixed rule. It reaches the leaderboard score of the 171 M ensemble we built for accuracy, while running 9.5 times faster and carrying 40 times fewer parameters. The whole path fits on one consumer GPU and needs 37 ms per image.

We make three claims. First, the accuracy limit of our earlier pipeline was its post-processing, not its model. Second, a 4.28 M student trained on every annotator and averaged over its last epochs is enough to tie a 171 M ensemble on the leaderboard. Third, the submitted path is deterministic and single pass, and it is released with the weights and a notebook that reproduce the score.

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

2 Method

Data and split. MAGFiLO holds 707 GONG H-α full-disk images at 2048×2048. There are 1,154 annotation records in total, between one and three annotators per image, and 8,199 instance masks. We split by image file, 600 for training and 107 for validation, so that the same image never appears in both sets through a different annotator. Every validation score in this paper is computed against all annotators of the validation images, using all-pairs IoU> 0.5 matching.

Preprocessing. Images are read as 8-bit grayscale. At inference we locate the solar disk on a 4× downsample with a threshold, a morphological close, a convex hull and an erosion. The disk is used only as a gate, and detections outside it are dropped. No external data enters this step, because the gate is computed from the test image itself. The single channel is replicated to three channels and normalized on the GPU with ImageNet statistics. We tested CLAHE and radial flattening, and both lowered validation PQ, so neither is used.

Model and training. The student is a ConvNeXt-atto encoder [5] with a four-stage U-Net decoder [6]. The decoder starts at 128 channels and the model has 4.28 M parameters. It is trained for 30 epochs on 1024-pixel crops at batch size 4, about 5 minutes per epoch on one GPU. The loss is 0.5 BCE + 0.5 Tversky with $\alpha = \beta = 0.5$. We use AdamW at 10^{-4}, a linear warmup followed by cosine decay, and a running average inside each batch. Inference uses the raw weights, because the averaged weights lose instance PQ. Training targets are the first annotator's mask.

A six-epoch fine-tune then moves to all 1,154 annotator records, so that successive steps see different annotators' versions of the same filament. It starts from the 30-epoch checkpoint, distils from a five-model teacher ensemble (MSE, weight 5) [3], and averages the weights of its last three epochs. Checkpoints during this fine-tune are selected by full-image PQ under the deployed rule, not by pixel Dice.

Inference and post-processing. Fig. 1 shows the path. The network is fully convolutional, so a whole 2048×2048 image goes through one forward pass in 37 ms at FP16. At submission time we average the sigmoid maps of eight dihedral views. The timings and figures in this paper use a single pass, because the same eight views are applied to every system we compare, so the ratio between systems does not change.

The probability map is then turned into instances by the KEEP rule. We take 8-connected components of $p > t_{\text{low}}$ and accept a

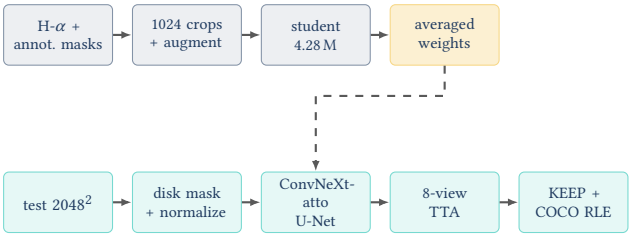


Figure 1: End-to-end pipeline. Navy marks the training path, which runs once. Teal marks the per-image inference path. The gold block is produced by training and initializes the network at inference.

component only if its peak exceeds t_{peak}, its mean exceeds t_{mean}, and its area is at least a_{min}. The submitted student uses $t_{\text{low}} = 0.35$, $t_{\text{peak}} = 0.99$, $t_{\text{mean}} = 0.85$ and $a_{\text{min}} = 100$. Accepted components are exported as COCO RLE.

3 Results

The three systems. The paper compares three systems. The *four-model ensemble* is our accuracy reference. It averages four ConvNeXt U-Nets, two tiny and two small, each reading the disk at 1024 pixels, for 171.0 M parameters in total. The *submitted student* is the 4.28 M ConvNeXt-atto network of Section 2. The *narrow student* is the same encoder with half the decoder width, 64 channels in its first stage, for 3.75 M parameters. It uses $t_{\text{low}} = 0.30$ and the same other thresholds. The five-model teacher ensemble that supervises the fine-tune is a separate object from the four-model accuracy reference.

Accuracy and speed. We use the competition metric, panoptic quality against every annotator with all-pairs IoU$>$ 0.5 matching,

$$\text{PQ} = \frac{\sum_{(p,g)\in TP} \text{IoU}(p,g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|} \tag{1}$$

[4]. Rule and ensemble choices are selected with nested 5-fold cross-validation on the 107 validation files and compared with a paired file-level bootstrap over 4,000 resamples.

Table 1 and Fig. 2 give the two numbers that matter. The submitted student reaches 0.4591 validation PQ and 0.38 on the public leaderboard. The ensemble reaches 0.4773 and also 0.38. The two are not equal on validation, where the ensemble is 0.018 ahead, but they are equal on the leaderboard, which reports two decimals. The student runs at 26.9 images per second against 2.8 for the ensemble, a factor of 9.5, at 40 times fewer parameters. The narrow student is faster still at 33.1 images per second and scores 0.0008 higher on validation, but it drops one leaderboard step to 0.37. That 0.0008 gap is much smaller than the bootstrap intervals we measure for other effects (Table 2), so we submitted the wider student and let the leaderboard decide.

Where the speed comes from. A full-disk image is four megapixels, and the student is fully convolutional, so there is no tiling, no patch stitching and no per-image parameter to set. One pass costs 37 ms. The KEEP rule and COCO RLE encoding add 34 ms, which gives 71 ms end to end, or 14 images per second, for a pipeline that returns

System	Params	img/s	Val PQ	Public LB
Four-model ensemble	171.0 M	2.8	**0.4773**	**0.38**
Submitted student	**4.28 M**	**26.9**	0.4591	**0.38**
Narrow student	3.75 M	33.1	0.4599	0.37

Table 1: Accuracy and throughput on the 107-file validation split. Throughput is in full 2048×2048 images per second on one RTX 4070, batch 1, FP16, over 30 validation images after warm-up. It counts one forward pass per model, so the ensemble row sums its four member passes. The eight-view augmentation used at submission applies to all three systems and is not included.

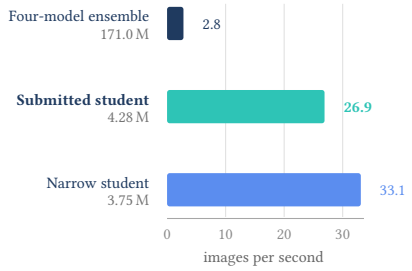


Figure 2: Throughput for one full-disk image. The submitted student is 9.5 times faster than the four-model ensemble at the same public leaderboard score of 0.38. The narrow student is faster again but scores 0.37. Parameter counts are printed under each name.

instances rather than a probability map. The earlier version of this pipeline ran an eight-crop 1024-pixel window, which needs 276 ms for the forward pass alone, so a single pass is about seven times cheaper for the same network. All timings are one RTX 4070 at batch 1 in FP16, averaged over 30 validation images after warm-up.

Error analysis. The remaining error is mostly under-detection. Under the deployed rule the student has 933 matched instances, 474 false positives and 461 false negatives. Of the false negatives, 292 (63.3%) are filaments the model never fires on, 163 (35.4%) are boundary disagreements with the annotator, and 6 (1.3%) are two filaments merged into one prediction. Matched pairs are close, with mean IoU 0.689 and mean Dice 0.813 over the 933 pairs (Fig. 4). Post-hoc dilation and erosion both lose PQ (dilation by one pixel −0.025, by two −0.094, erosion by one −0.022), so what remains is not a thin-boundary problem.

Fragmentation. Under IoU$>$ 0.5 matching the student has no one-to-many and no many-to-one relations. That is a property of the matching, which gives each instance at most one partner, and not a separate result. The informative count is at any overlap. At IoU$>$ 0, 2.4% of ground-truth instances overlap more than one prediction and 5.1% of predictions overlap more than one ground-truth instance (Fig. 4, lower panel). The KEEP rule does not act on those duplicates. It acts on false positives: moving from the older coupled rule to KEEP changed TP, FP and FN from 916/575/478 to 933/474/461, adding 17 true positives and removing 101 false

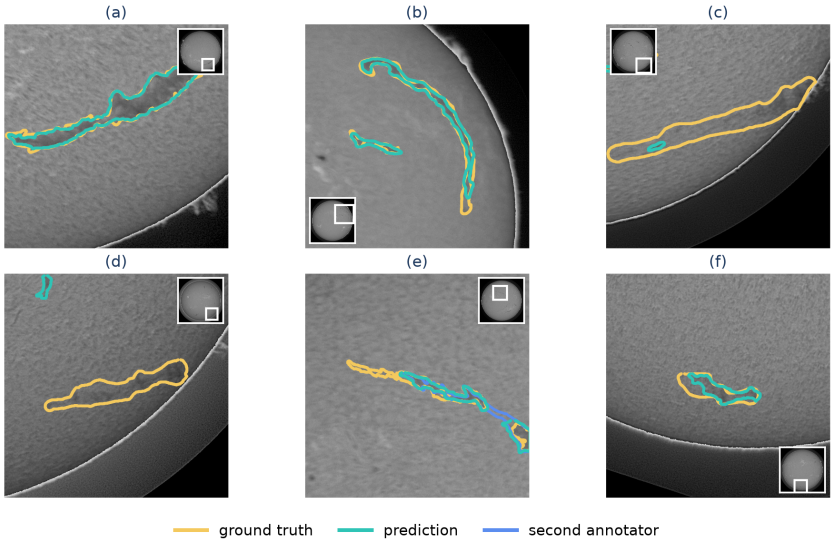
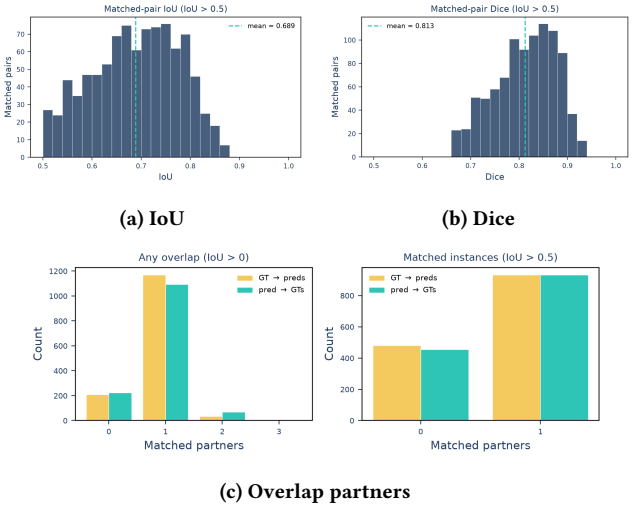


Figure 3: Zoomed validation examples. Gold is the ground truth, teal the prediction, blue a second annotator. In (a) and (b) the prediction follows thin barbs and a large curved filament. In (c) and (d) two filaments are missed outright. In (e) and (f) the annotators disagree and the prediction follows one of them. Insets locate each crop on the disk.



(a) IoU **(b) Dice**

(c) Overlap partners

Figure 4: Matched-pair overlap and partner counts for the submitted student on the 107 validation files. The lower panel counts partners at IoU>0 (left) and under the official IoU>0.5 matching (right). Under IoU>0.5 every instance has zero or one partner, so duplicate relations cannot reach the metric.

positives. Recognition quality rose from 0.635 to 0.666 while segmentation quality moved only from 0.686 to 0.689.

Qualitative comparison. Fig. 3 shows zoomed validation crops. Panels (a) and (b) contain the behavior we want, where thin barbs and curved bodies are kept. Panels (c) and (d) contain the failure we keep seeing, a filament that produces no response at all. Panels (e) and (f) contain cases where the annotators disagree and the prediction follows one of them. Limb artifacts do not appear, because the disk gate removes them before the KEEP rule reads the map.

Effect (nested out-of-fold)	ΔPQ	95% CI
KEEP rule vs. coupled rule	+0.0158	[0.0087, 0.0231]
ConvNeXt-tiny, annotator sampling	+0.0127	[0.0045, 0.0217]
ConvNeXt-tiny, sampling + averaging	+0.0105	[0.0032, 0.0177]
Submitted student, sampling + averaging	+0.0051	[−0.0056, 0.0167]

Table 2: Paired file-level bootstrap over 4,000 resamples for the rule and fine-tune gains. All intervals are nested 5-fold out-of-fold on the 107 validation files. Each row is a gain over the same model without that change.

What the rule and the fine-tunes are worth. Table 2 collects the effects we measured with nested 5-fold cross-validation on the 107 validation files and a paired file-level bootstrap. The KEEP rule is worth 0.0158 PQ against the older coupled rule, and its interval is clear of zero. The annotator-sampling fine-tune is worth 0.0127 on a ConvNeXt-tiny student, and 0.0105 once weight averaging is added. On the submitted student the same fine-tune is worth 0.0051, and its interval includes zero, so we do not claim it as a gain. The rule is the one change that moved the leaderboard on its own.

A better member does not help the ensemble. Adding the fine-tuned student to the four-model ensemble at weights of 0.10, 0.15 and 0.20 moves PQ by +0.0003, −0.0001 and +0.0002, and every interval spans zero. The same member added at weight 0.10 gives +0.0001 (−0.0024 to +0.0026). A member can be clearly better on its own and still add nothing, because all members miss the same unlabeled filaments, so the error that remains is shared rather than

averaged away. That is why we dropped the ensemble from the submission.

Limitations. Three limits bound these numbers. Validation labels are noisy, with an inter-annotator union-mask IoU of 0.467, and 27% of our confident detections match no annotator at all, so absolute validation PQ is optimistic and only deltas travel. The public leaderboard is rounded to 0.01 and cannot resolve the effects in Table 2, which is why we select the accuracy reference on validation PQ and report intervals rather than absolute levels alone. The largest untested lever is data coverage, namely five-fold training over all 707 images instead of 600, rather than a different architecture.

Reproducibility. The released package contains the 16.4 MB student weight, the narrow student weight, an inference notebook, a 13-check submission verifier and the two timing scripts behind Fig. 2. The notebook reproduces 0.4591 on the same 107 files. Re-inference over the test set takes about a minute on one GPU and reproduces 1,277 of the 1,279 submitted rows byte for byte, with the one remaining image shifting two contours by a pixel and adding a boundary component that clears the area test.

4 Conclusion

We submitted a 4.28 M-parameter network that processes a full-disk H-α image in 37 ms and scores 0.38 on the public leaderboard, the same as a 171 M four-model ensemble that is 9.5 times slower. Most of the accuracy we found was in post-processing. Letting the contour threshold and the instance-acceptance test be set independently was worth 0.0158 PQ and one leaderboard step on its own, with no retraining. Training on every annotator and averaging the last epochs gave a smaller and less certain gain. What remains is under-detection on faint, low-contrast filaments, which points to more training data rather than a different architecture.

5 Acknowledgment

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References

[1] Azim Ahmadzadeh, Rohan Adhyapak, Kartik Chaurasiya, Laxmi Alekhya Nagubandi, V. Aparna, Petrus C. Martens, Alexei A. Pevtsov, Luca Bertello, Naomi Douglas, Samuel McDonald, Apaar Bawa, Eugene Kang, Riley Wu, Dustin J. Kempton, Aya Abdelkarem, Patrick M. Copeland, and Sri Harsha Seelamneni. 2024. A dataset of manually annotated filaments from H-alpha observations. *Scientific Data* 11, 1 (2024), 1031. doi:10.1038/s41597-024-03876-y

[2] Azim Ahmadzadeh, Dustin J. Kempton, Qin Li, and Alexei A. Pevtsov. 2026. Solar Filament Segmentation Challenge 2026. https://kaggle.com/competitions/filament-segmentation-2026. Kaggle.

[3] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531* (2015).

[4] Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. 2019. Panoptic Segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9404–9413.

[5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. 2022. A ConvNet for the 2020s. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11976–11986.

[6] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 234–241.